

✓ Key Concepts in Programming

1. Variable
2. Type
3. Structure
4. Control Flow
5. Function

```
1 ## special topics in Python
```

```
1 ## type hint
2
3 name: str = "Toy"
4 age: int = 37
5 gpa: float = 3.41
6 netflix: bool = True
```

```
1 type(age), type(gpa), type(name), type(netflix)
```

```
➦ (int, float, str, bool)
```

```
1 ## type hint in function
2 ## optional
3 def add(a: int, b: int) -> int:
4     return a + b
5
6 add(5,7)
```

```
➦ 12
```

```
1 from typing import List
```

```
1 friends: List[str] = ["toy", "john", "mary"]
2
3 for f in friends:
4     print(f"Hi! {f}")
```

```
➦ Hi! toy
  Hi! john
  Hi! mary
```

```
1 ## lambda one-liner function
2 add_lambda = lambda a,b : a+b
3
4 add_lambda(1, 2)
```

```
➦ 3
```

```
1 ## list comprehension
2 result = []
3 original = [1,2,3,4,5]
4
5 for number in original:
6     result.append(number * 2)
7
8 print(result)
```

```
➦ [2, 4, 6, 8, 10]
```

```
1 [number*2 for number in original]
```

```
➦ [2, 4, 6, 8, 10]
```

```
1 ## grade
2 ## score >= 80, passed or failed
3 scores = [88, 75, 72, 90, 95]
4
```

```

5 ["passed" if score >= 80 else "failed"
6   for score in scores]

```

```

→ ['passed', 'failed', 'failed', 'passed', 'passed']

```

```

1 for score in scores:
2     if score >= 80:
3         print("passed")
4     else:
5         print("failed")

```

```

→ passed
failed
failed
passed
passed

```

```

1 ## try - except working with error
2
3 try:
4     result = 1/0
5     print(result)
6 except:
7     print("cannot divide by zero")

```

```

→ cannot divide by zero

```

```

1 ## try print a variable
2 datarockie = 500
3
4 try:
5     print(int("hello"))
6 except NameError:
7     print("no variable called 'datarockie'")
8 except ZeroDivisionError:
9     print("cannot divide by zero")
10 except ValueError:
11     print("check your value")
12 else:
13     print("the value is printed")
14 finally:
15     print("the end!")

```

```

→ check your value
the end!

```

```

1 ## how to read csv file in python
2 import csv

```

```

1 !cat hotel.csv

```

```

→ id,hotel,price,location
1,Santa Cruz,500,Bangkok
2,Terminal,1500,Seoul
3,Grand Plaza,2000,Tokyo
4,The Mall,1800,London

```

```

1 import csv

```

```

1 # read file
2 file = open("hotel.csv", "r")

```

```

1 # read content
2 data = csv.reader(file)
3
4 for row in data:
5     print(row)

```

```

→ ['\ufeffid', 'hotel', 'price', 'location']
['1', 'Santa Cruz', '500', 'Bangkok']
['2', 'Terminal', '1500', 'Seoul']
['3', 'Grand Plaza', '2000', 'Tokyo']
['4', 'The Mall', '1800', 'London']

```

```
1 ## close file
2 file.close()
```

```
1 ## with : context manager
2 with open("hotel.csv", "r") as file:
3     data = csv.reader(file)
4     print([row for row in data])
```

```
[[['\uffeffid', 'hotel', 'price', 'location'], ['1', 'Santa Cruz', '500', 'Bangkok'], ['2', 'Terminal', '1500', 'Seoul'], ['3', 'Grand Pla
```

```
1 ## write csv file
2 import csv
3
4 header = ["id", "student_name", "age"]
5 body = [
6     [1, "toy", 25],
7     [2, "ann", 22],
8     [3, "joe", 28]
9 ]
10
11 with open("new_file.csv", "w") as file:
12     writer = csv.writer(file)
13     writer.writerow(header)
14     writer.writerows(body)
```

```
1 !cat new_file.csv
```

```
id,student_name,age
1,toy,25
2,ann,22
3,joe,28
```

```
1 ## with + try
2 try:
3     with open("new_files.csv", "r") as file:
4         data = csv.reader(file)
5         for row in data:
6             print(row)
7 except:
8     print("file not found")
```

```
file not found
```

```
1 !ls
```

```
hotel.csv new_file.csv sample_data
```

```
1 # sklearn basics
2 # ref: https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LinearRegression.html
3 from sklearn.linear_model import LinearRegression
4 from sklearn.ensemble import RandomForestRegressor
5 from sklearn.tree import DecisionTreeRegressor
6 from sklearn.model_selection import train_test_split
7 import pandas as pd
8
9 # read data
10 mtcars = pd.read_csv("https://gist.githubusercontent.com/seankross/a412dfbd88b3db70b74b/raw/5f23f993cd87c283ce766e7ac6b329ee7cc2e1d1/mtc
11 mtcars.head()
```

```

model mpg cyl disp hp drat wt qsec vs am gear carb
0 Mazda RX4 21.0 6 160.0 110 3.90 2.620 16.46 0 1 4 4
1 Mazda RX4 Wag 21.0 6 160.0 110 3.90 2.875 17.02 0 1 4 4
2 Datsun 710 22.8 4 108.0 93 3.85 2.320 18.61 1 1 4 1
3 Hornet 4 Drive 21.4 6 258.0 110 3.08 3.215 19.44 1 0 3 1
4 Hornet Sportabout 18.7 8 360.0 175 3.15 3.440 17.02 0 0 3 2
```

Next steps:

[Generate code with mtcars](#)

[View recommended plots](#)

[New interactive sheet](#)

```
1 # prepare data
2 X = mtcars.drop(["model", "mpg"], axis=1)
3 y = mtcars["mpg"]
4
5 # split data
6 # set.seed(42)
7 X_train, X_test, y_train, y_test = train_test_split(
8     X, y, test_size = 0.25, random_state = 24
9 )
10
11 # train model
12 model = RandomForestRegressor()
13 model.fit(X_train, y_train)
14
15 # prediction
16 p = model.predict(X_test)
17
18 # R squared
19 model.score(X_test, y_test)
```

 0.867064230990655

1 Start coding or [generate](#) with AI.